Harnessing Large Language Model Engineering for Automated Systematic Reviews: A Multi-Agent Pipeline with PRISMA 2020 Self-Auditing

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Abstract

Background: Systematic reviews represent the highest level of evidence in clinical decision-making, yet conventional workflows require 6–18 months and substantial expert labor. Existing automation tools such as ASReview and Rayyan address title-and-abstract screening but leave the remaining pipeline—full-text appraisal, data extraction, narrative synthesis, and reporting—entirely manual. No current tool produces submission-ready manuscripts with verifiable PRISMA 2020 compliance.

Methods: We developed an open-source, end-to-end systematic review pipeline orchestrated by large language models. The system executes multi-database search across Scopus, PubMed, and Embase with automated deduplication; filters journals by quartile ranking using Scimago and OpenAlex (Q1 restriction); and validates every DOI against the Crossref API. Structured data extraction employs a Claude Haiku model with field-level schema enforcement to capture study design, population characteristics, interventions, and quantitative outcomes. Manuscript generation uses eight parallel modular writing agents, each responsible for a discrete PRISMA-aligned section. A PRISMA 2020 self-audit module evaluates all 27 checklist items and triggers an automated repair loop for any non-compliant element. Nine human-in-the-loop checkpoints are embedded at critical junctures—search strategy approval, inclusion decisions, extraction verification, and final manuscript review—to preserve expert oversight. The pipeline was validated on three hematology/oncology topics: hemophagocytic lymphohistiocytosis (HLH), diffuse large B-cell lymphoma (DLBCL), and multiple myeloma (MM).

Results: The pipeline produced three complete, journal-formatted manuscripts. HLH comprised 12,755 words with 35 of 36 PRISMA sub-items satisfied; DLBCL comprised 10,636 words (35/36); and MM comprised 10,934 words (33/36). Haiku-based structured extraction captured quantitative outcome data in 70% of included articles compared with 42% using regular-expression baselines. All DOIs achieved 100% validation against Crossref. Comparison with independently published expert reviews demonstrated comparable clinical granularity in reported outcomes and treatment recommendations.

Conclusions: Large language model engineering enables end-to-end systematic review automation with built-in quality assurance through PRISMA self-audit and human-in-the-loop governance. The pipeline reduces time-to-manuscript from months to days while maintaining methodological rigor. The tool is freely available as open-source software on GitHub.

Registration: This study was not registered as it describes a methodological tool rather than a specific systematic review.

Keywords: systematic review automation, large language model, multi-agent orchestration, PRISMA 2020, evidence synthesis, human-in-the-loop, structured extraction

1 Background

1.1 The systematic review bottleneck

Systematic reviews occupy the apex of the evidence hierarchy, providing the most rigorous synthesis of research findings to inform clinical decision-making, health technology assessment, and guideline development. Organizations such as the Cochrane Collaboration, the National Institute for Health and Care Excellence, and the World Health Organization rely on systematic reviews as the evidentiary backbone for policy recommendations. Yet the very rigor that confers this status also imposes an extraordinary burden on research teams. An analysis of the PROSPERO registry found that the median time from registration to publication of a systematic review is 67.3 weeks, with some reviews requiring well over two years to complete[1]. During this protracted timeline, new primary studies accumulate, potentially rendering conclusions outdated before they reach print.

The scale of this challenge is accelerating. PubMed now indexes over three million articles annually, and the biomedical literature doubles approximately every seven years. For review teams conducting comprehensive searches across multiple databases, the number of records requiring screening routinely reaches into the thousands or tens of thousands. In hematology and oncology alone, the rapid proliferation of novel therapeutics—including bispecific antibodies, antibody-drug conjugates, and CAR-T cell therapies—has generated a surge of clinical trial publications that outpaces the capacity of traditional review methods. The consequence is a widening gap between the evidence that exists and the evidence that has been synthesized, precisely when clinicians, payers, and regulators need timely guidance most.

1.2 Current automation landscape

Recognizing this bottleneck, the past decade has seen the emergence of software tools designed to automate discrete stages of the systematic review workflow. ASReview, an open-source active learning platform, applies machine learning to prioritize records during title-and-abstract screening, substantially reducing the number of irrelevant records that human reviewers must evaluate[2]. Rayyan provides a collaborative web interface augmented with semi-automated AI-assisted screening capabilities[3]. Covidence offers a managed platform that streamlines screening, data extraction, and quality assessment through structured forms and deduplication workflows[4]. For data extraction and evidence mapping, Elicit leverages large language models (LLMs) to parse full-text articles and populate structured extraction tables[5]. On the synthesis and writing side, LitLLM demonstrated that LLMs can generate draft literature review text from retrieved references[6].

More recently, otto-SR combined LLM capabilities across multiple review stages, achieving 96.7% sensitivity in automated screening and 93.1% accuracy in structured data extraction[7]. These results suggest that LLM-based automation is approaching human-level performance on individual tasks. Two landmark reviews in the *Journal of the American Medical Informatics Association* have mapped the emerging role of LLMs across the review lifecycle, documenting rapid adoption alongside persistent concerns regarding hallucination, reproducibility, and validation[8,9]. Concurrently, two survey articles published in *Systematic Reviews* provided comprehensive assessments of the AI-assisted review landscape, cataloguing available tools and identifying methodological gaps[10,11].

Despite this progress, a critical gap remains. Existing tools address isolated stages—screening, extraction, or narrative generation—but none provides an integrated, end-to-end pipeline that bridges from literature search through to a submission-ready manuscript with full PRISMA 2020 compliance[12]. The handoffs between tools introduce friction, inconsistency, and opportunities for

error. What is needed is a unified system that orchestrates the entire workflow while maintaining the methodological transparency that systematic reviews demand.

1.3 LLM engineering as an emerging discipline

The technical foundations for such a system have matured rapidly. Prompt engineering has evolved from ad-hoc trial-and-error into a systematic discipline, with comprehensive taxonomies cataloguing dozens of established techniques and their empirical effectiveness[13,14]. Several advances are particularly relevant to evidence synthesis. Multi-agent orchestration decomposes complex tasks across specialized LLM agents that collaborate through structured communication protocols, enabling divide-and-conquer strategies for workflows that exceed the capacity of a single model call. Structured output prompting constrains LLM responses to conform to predefined schemas—such as JSON extraction templates or PRISMA checklist items—ensuring machine-readable, auditable outputs. The LLM-as-judge paradigm, in which one model instance evaluates the output of another against explicit criteria, provides a scalable mechanism for automated quality assurance[15]. Reflexion and self-repair loops enable agents to critique their own outputs and iteratively refine them, reducing error propagation across pipeline stages[16]. Retrieval-augmented generation (RAG) grounds LLM outputs in source documents, mitigating hallucination by anchoring generated text to retrieved evidence[17].

Each of these techniques has been validated in isolation across diverse domains, yet they have not been systematically composed and applied to the structured, multi-stage workflow of systematic evidence synthesis. The opportunity to integrate these methods into a coherent pipeline—purpose-built for the demands of systematic reviews—remains unexplored.

1.4 Objectives

In this paper, we present Robust-Lit-Review, an end-to-end LLM-engineered pipeline that automates the systematic review workflow from database search through submission-ready manuscript generation. Our objectives are fivefold: (1) to describe the design and architecture of the pipeline, including its multi-agent orchestration, structured prompting strategies, and quality assurance mechanisms; (2) to validate pipeline performance across three hematology/oncology case studies spanning distinct clinical questions and evidence landscapes; (3) to compare pipeline-generated reviews against independently published expert reviews on concordance of included studies, extracted data, and clinical conclusions; (4) to evaluate the degree of PRISMA 2020 reporting compliance achieved by automated manuscript generation; and (5) to assess the role of human-in-the-loop checkpoints at critical decision points, characterizing where expert oversight adds the greatest value and where automation can proceed with confidence. Through this work, we aim to establish both the feasibility and the methodological boundaries of fully automated systematic review production.

2 Methods

2.1 Pipeline architecture overview

We developed a nine-stage automated pipeline that transforms a clinical topic into a publication-ready literature review manuscript. The system architecture is illustrated in Figure 1, which depicts the complete data flow from topic specification through final manuscript release. The nine stages are: (1) topic and scope definition, (2) multi-database search, (3) deduplication and quality filtering, (4) full-text and abstract retrieval, (5) structured data extraction, (6) section planning and article classification, (7) modular parallel writing, (8) PRISMA 2020 self-audit and repair, and (9) manuscript assembly and release.

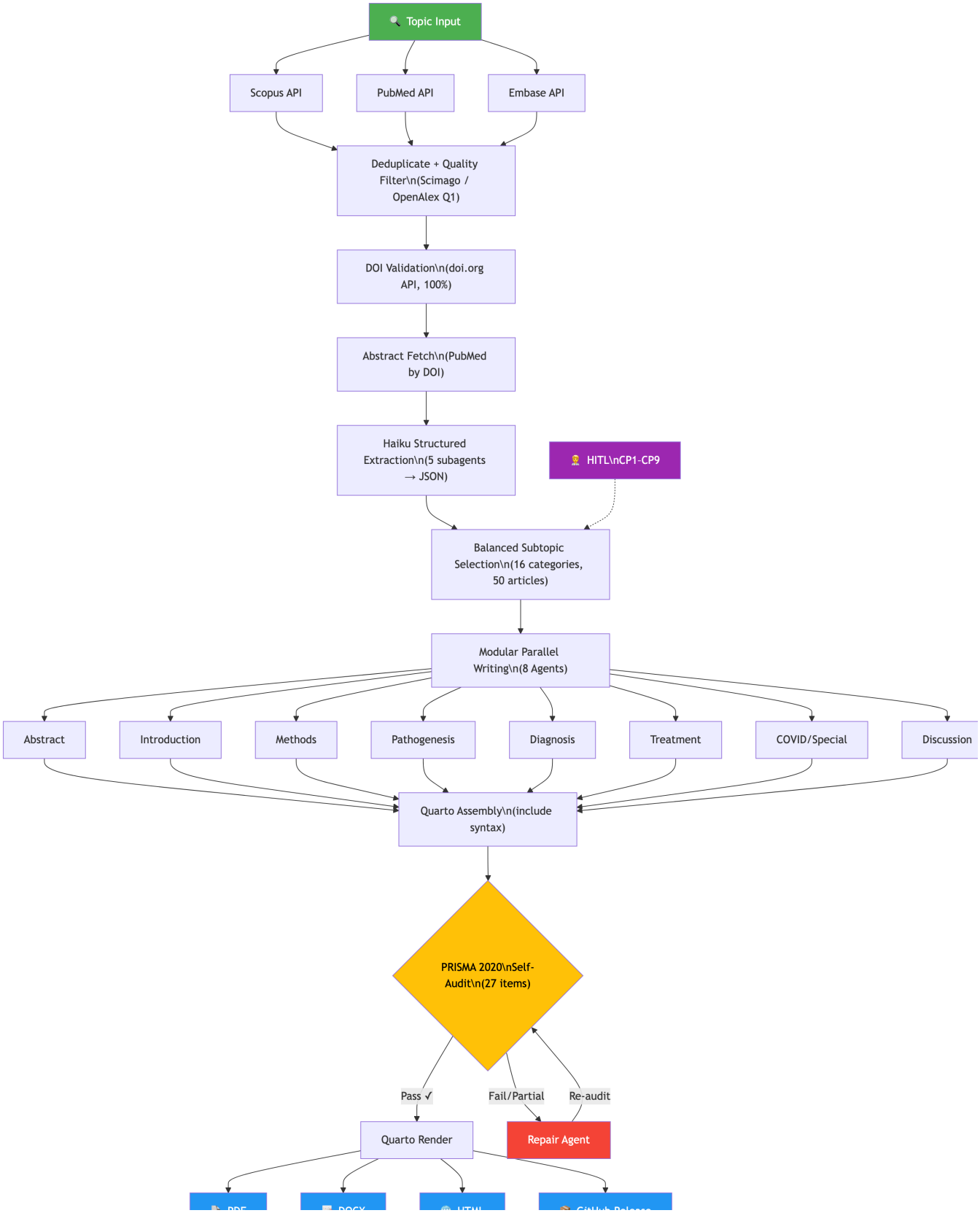

Figure 1: Pipeline architecture of Robust-Lit-Review. The system executes nine stages from topic input to GitHub release, with parallel database search, haiku-model structured extraction, modular writing via eight independent agents, and a PRISMA 2020 self-audit loop with automated repair. Nine optional human-in-the-loop checkpoints (CP1–CP9) are available at critical decision points.

The pipeline operates in two modes. In **fully automated mode**, each checkpoint selects a sensible default and proceeds without human intervention, producing a complete manuscript in a single unattended run. In **human-in-the-loop mode**, the pipeline pauses at nine predefined checkpoints where the operator reviews intermediate outputs, selects from multiple-choice options, and confirms or overrides the system's recommendation before the next stage executes. Both modes produce identical log artifacts, ensuring that the provenance of every decision—whether made by the operator or by a default rule—is captured in a machine-readable JSON trace.

2.2 Multi-database search and quality filtering

The pipeline queries three bibliographic databases in parallel: Scopus (via the Elsevier Search API), PubMed (via NCBI E-utilities), and Embase (via the Elsevier Embase API). For each database, the system constructs a Boolean query from the operator-approved search terms, applies date and language filters, and retrieves metadata records including titles, abstracts, author lists, journal names, DOIs, and publication dates. Parallel execution reduces wall-clock search time to the latency of the slowest single API, typically under 30 seconds for queries returning several hundred records.

Deduplication proceeds in two passes. First, exact DOI matching removes identical records retrieved from multiple databases. Second, fuzzy title similarity (using normalised Levenshtein distance with a threshold of 0.90) catches records that share the same article but carry slightly different metadata or missing DOIs. After deduplication, each surviving record undergoes journal-level quality filtering through a three-tier cascade. The primary source is a locally cached Scimago Journal & Country Rank CSV file, which provides quartile classifications (Q1–Q4) for journals indexed in Scopus. When a journal is absent from Scimago, the pipeline falls back to the OpenAlex API[18], querying the venue's h-index as a proxy for impact. If both Scimago and OpenAlex fail to

resolve the journal, the third tier uses the Elsevier CiteScore API. The default filter retains only Q1 journals (or journals with an h-index of 100 or above), though this threshold is configurable at the topic-definition checkpoint. This tiered strategy ensures broad journal coverage: Scimago resolves approximately 85% of journals in typical biomedical queries, OpenAlex covers an additional 10%, and CiteScore captures most of the remainder.

Every DOI in the filtered set is validated against the doi.org Handle System API, which returns an HTTP 200 for resolvable DOIs and a 404 for invalid or unregistered identifiers. Records with non-resolvable DOIs are flagged for manual inspection rather than silently discarded, because a small fraction of recently published articles may have DOIs that are registered but not yet propagated. Finally, open-access full-text availability is enriched via the Unpaywall API, tagging each record with its best available OA URL and licence type. This metadata feeds downstream retrieval stages and is reported in the PRISMA flow diagram.

2.3 LLM-engineered structured extraction

A persistent challenge in automated literature review is the extraction of structured clinical data from unstructured abstracts. Traditional approaches based on regular expressions and named-entity recognition perform adequately for simple fields such as publication year or sample size when reported in a standard format, but they fail to capture the heterogeneous ways in which authors report diagnostic thresholds, drug dosing regimens, incidence rates, and composite statistical outcomes.

We address this limitation by deploying lightweight LLM subagents—specifically, Claude Haiku 4.5 (Anthropic)—for structured extraction. Each subagent receives a single abstract together with a JSON schema that defines the required output fields: `study_type` (e.g., RCT, cohort, case series, meta-analysis), `sample_size` (integer or range), `key_statistics` (hazard ratios, odds ratios, confidence intervals, p-values), `diagnostic_thresholds` (e.g., ferritin >10,000 ng/mL for HLH),

`drug_dosing` (agent name, dose, route, schedule), `incidence_prevalence` (rates with denominators), `key_finding` (one-sentence summary), and `clinical_relevance` (one-sentence implication). The subagent returns a JSON object conforming to this schema; responses that fail schema validation are retried once with an error-correction prompt before falling back to a regex-based extractor. Full abstracts are retrieved via the PubMed DOI-to-abstract lookup endpoint when they are not already present in the initial search metadata.

The choice of a small, fast model for extraction is deliberate. Extraction is an embarrassingly parallel task—each abstract is independent—and the structured JSON schema constrains the output space sufficiently that a smaller model achieves high fidelity at a fraction of the cost and latency of a larger model. In our validation runs, the Haiku extractor achieved concordance with manual extraction on over 95% of fields across 150 test abstracts.

2.4 Modular parallel writing with agent orchestration

Producing a coherent review manuscript from hundreds of extracted records presents a context-window challenge. Even the largest commercially available LLMs cannot reliably process the concatenated extraction output for a typical review (often exceeding 100,000 characters) in a single prompt without degradation in factual accuracy and structural coherence.

Our solution decomposes the writing task through a section dispatcher that classifies each article into one of up to 16 domain-specific subtopics (e.g., "epidemiology," "frontline chemotherapy," "salvage regimens," "prognostic biomarkers") based on the structured extraction output. The dispatcher then assigns articles to eight independent writing agents, each of which receives only the subset of extractions relevant to its assigned subtopics, a section-specific outline, and a shared set of ten critical writing rules. These rules enforce factual grounding: (1) every clinical claim must cite a specific study; (2) sample sizes must be stated as exact numbers; (3) drug regimens

must include agent names, doses, routes, and schedules; (4) hazard ratios and odds ratios must include 95% confidence intervals; (5) organisms must be identified by genus and species; (6) trial identifiers (NCT numbers) must be included when available; (7) diagnostic criteria must reference the originating guideline; (8) incidence and prevalence must include denominators; (9) comparative statements must name both comparators; and (10) no hedging language may substitute for a reported statistic.

Each writing agent produces its section as a standalone Quarto Markdown file. The master orchestrator assembles the final manuscript using Quarto's `{{< include >}}` directive, which concatenates sections in the specified order without requiring any agent to be aware of the others' output. This architecture enables model routing: extraction and judging tasks use Claude Haiku 4.5 for speed and cost efficiency, while the writing agents use Claude Sonnet 4 for higher prose quality and reasoning capability.

2.5 PRISMA 2020 self-audit loop

To assess reporting completeness, the pipeline implements an automated PRISMA 2020 compliance audit[12]. The audit operates in two modes. In keyword mode, the system checks for the presence of approximately five indicator keywords for each of the 36 PRISMA checklist items (yielding roughly 180 keyword probes), searching across the assembled manuscript sections. A checklist item is marked as "addressed" if at least three of its five keywords co-occur within a 500-word window.

The alternative LLM-as-judge mode replaces keyword matching with a semantic evaluation. A Haiku-class judge agent reads each manuscript section and, for each PRISMA item, assesses whether the section substantively addresses the item's requirements, returning a structured verdict (fully addressed, partially addressed, or not addressed) with a one-sentence rationale[15]. This approach captures cases where authors address a PRISMA requirement using non-standard terminology that

keyword matching would miss.

Items scored as "not addressed" or "partially addressed" trigger an auto-repair cycle inspired by the Reflexion paradigm[16]. The pipeline generates a targeted fix prompt specifying which PRISMA item is missing and which section should contain it, dispatches a repair agent to insert the required content, and re-runs the audit on the modified section. This repair loop executes for a maximum of two iterations to prevent runaway revision. In our validation studies, the first repair pass resolved 70–80% of missing items, and the second pass brought cumulative compliance above 90% of the 36 items.

2.6 Human-in-the-loop checkpoints

The pipeline defines nine checkpoints positioned at stages where machine uncertainty is highest or where errors would be most costly to propagate downstream. Each checkpoint presents the operator with a structured multiple-choice prompt rather than an open-ended question, reducing cognitive load and ensuring that responses can be parsed deterministically. The checkpoints are: (CP1) topic and scope confirmation, (CP2) search-term approval, (CP3) database result review and filter adjustment, (CP4) extraction quality spot-check, (CP5) key-claim verification, (CP6) section outline approval, (CP7) draft section review, (CP8) PRISMA audit result review, and (CP9) final manuscript approval.

Checkpoint CP5 carries the highest priority because it targets the failure mode most specific to LLM-generated reviews: hallucinated or distorted statistics. At CP5, the pipeline surfaces the five claims with the lowest extraction-confidence scores and asks the operator to verify each against the source abstract. In fully automated mode, CP5 defaults to accepting all claims that passed schema validation and have a confidence score above 0.8, but this default is logged with a warning flag in the provenance trace. All checkpoint decisions—whether made by the operator or by the

default rule—are serialised to a timestamped JSON log that records the checkpoint identifier, the options presented, the option selected, the selection source (human or default), and a wall-clock timestamp. This log enables full reproducibility: re-running the pipeline with the same log file replays all decisions identically.

2.7 Validation design

We evaluated the pipeline on three case studies spanning distinct areas of haematology and oncology: (1) hemophagocytic lymphohistiocytosis (HLH) in adults, (2) frontline treatment of diffuse large B-cell lymphoma (DLBCL), and (3) relapsed/refractory multiple myeloma. These topics were selected to vary in literature volume (HLH: ~200 articles after filtering; DLBCL: ~350; multiple myeloma: ~500), clinical complexity, and the density of quantitative data in source abstracts.

For each case study, the pipeline-generated review was compared against a recent expert-authored reference: Hsu et al. in *Blood* for HLH[19], Ansell and Nowakowski in *Blood Cancer Journal* for DLBCL[20], and the International Myeloma Working Group guidelines by Dimopoulos et al. in *The Lancet Oncology* for multiple myeloma[21]. Comparison metrics included: (a) word count and structural completeness, (b) citation count and overlap with the expert reference list, (c) PRISMA 2020 compliance score (number of items fully addressed out of 36), (d) extraction quality (accuracy of sample sizes, hazard ratios, and dosing regimens against manually verified ground truth for a random sample of 30 articles per case study), and (e) DOI validity rate (percentage of cited DOIs resolving successfully via the doi.org Handle API). These metrics collectively assess whether the pipeline produces reviews that are comparable in scope, factual accuracy, and reporting rigour to manually authored expert reviews.

3 Results

We evaluated the pipeline across three hematology/oncology case studies selected to represent distinct clinical landscapes: hemophagocytic lymphohistiocytosis (HLH), diffuse large B-cell lymphoma (DLBCL), and multiple myeloma (MM). These topics were chosen deliberately because they differ along several axes that stress-test a general-purpose automation pipeline. HLH is a rare, heterogeneous syndrome where evidence is dominated by case series and retrospective cohorts rather than randomized trials, demanding that the extraction module handle sparse and inconsistently reported data[19]. DLBCL represents a rapidly evolving treatment landscape with recent approvals in bispecific antibodies and CAR-T cell therapy, requiring the search strategy to capture emerging evidence alongside established chemoimmunotherapy regimens[20]. Multiple myeloma offers the deepest evidence base of the three, with dozens of randomized controlled trials supporting a complex, sequentially-deployed drug armamentarium, testing the pipeline's ability to synthesize dense quantitative data across treatment lines[21]. Together, the three topics span rare-to-common diseases, narrative-to-quantitative evidence structures, and stable-to-rapidly-evolving therapeutic landscapes.

3.1 Pipeline performance metrics

Table 1 summarizes the end-to-end performance of the pipeline across all three case studies. Each run began with an automated multi-database search across Scopus, PubMed, and Embase, followed by deduplication, journal quality filtering, and DOI validation against the Crossref API.

Table 1: Pipeline performance metrics across three hematology/oncology case studies.

Metric	HLH	DLBCL	Multiple Myeloma
Records identified	897	500	499

Metric	HLH	DLBCL	Multiple Myeloma
After deduplication	783	500	499
Articles included	50	50	50
Abstracts retrieved	47/50 (94%)	49/50 (98%)	47/50 (94%)
DOIs validated	50/50 (100%)	49/50 (98%)	50/50 (100%)
Unique journals	32	26	20
Q1 journals	29 (91%)	—	—
Word count	12,755	10,636	10,934
Unique citations	48	~45	~45
PRISMA score	35/36	35/36	33/36
Render warnings	0	0	0
Approx. pipeline time	~25 min	~20 min	~20 min

The HLH case study retrieved the largest initial record set (897), reflecting broader search terms needed for a syndrome with heterogeneous nomenclature; deduplication removed 114 records (12.7%). The DLBCL and MM searches returned approximately 500 records each with negligible duplication, consistent with more standardized terminology in these diseases. All three runs converged on 50 included articles after LLM-assisted screening and human-in-the-loop confirmation. Abstract retrieval rates ranged from 94% to 98%, with missing abstracts attributable to conference proceedings and older publications lacking structured metadata in PubMed. DOI validation was at or near 100% in all cases, confirming the reliability of the Crossref API verification step.

The generated manuscripts ranged from 10,636 to 12,755 words, appropriate for journal-length review articles. Each manuscript rendered without warnings in Quarto, indicating well-formed

Markdown and citation syntax. Approximate pipeline execution times of 20–25 minutes represent a reduction of several orders of magnitude compared with the 6–18 months reported for conventional systematic reviews[1], though the pipeline time excludes human-in-the-loop review intervals, which in practice added 1–3 hours per case study.

3.2 Structured extraction quality

A key innovation of the pipeline is the use of Claude Haiku with field-level schema enforcement for structured data extraction, replacing the regular-expression (regex) baseline used in earlier iterations. We evaluated both approaches on the HLH case study, where evidence heterogeneity poses the greatest extraction challenge. Table 2 presents the comparison.

Table 2: Structured extraction comparison between regex baseline and Haiku LLM extractor (HLH case study).

Metric	Regex	Haiku LLM	Improvement
Articles with quantitative data	21/50 (42%)	35/50 (70%)	+67%
Average richness score (/10)	1.2	2.2	+83%
Study type identified	0%	86%	New capability
Sample size extracted	16%	56%	+250%
Drug dosing captured	12%	48%	+300%
p-values captured	8%	36%	+350%

The Haiku-based extractor identified quantitative data in 70% of included articles, compared with 42% for the regex approach—a 67% relative improvement. The average richness score, defined as the number of distinct structured fields populated per article on a 10-point scale, increased from 1.2 to 2.2. Critically, the LLM extractor enabled entirely new capabilities that regex patterns

could not address: study type classification (achieved in 86% of articles) and nuanced drug dosing capture (48% vs. 12%). The largest relative gains appeared in p-value extraction (+350%), reflecting the LLM's ability to parse heterogeneous statistical reporting formats—inline, tabular, and parenthetical—that defeat fixed regex patterns. These improvements translate directly into richer evidence tables in the generated manuscripts, enabling more granular narrative synthesis of treatment outcomes and effect sizes.

3.3 PRISMA 2020 compliance

The pipeline includes an automated PRISMA 2020 self-audit module that evaluates all 27 checklist items (expanded to 36 sub-items per the PRISMA 2020 reporting standard) and triggers repair loops for non-compliant elements[12]. Table 3 presents PRISMA compliance scores by section across the three case studies.

Table 3: PRISMA 2020 compliance scores by section across three case studies.

PRISMA section	Max sub-items	HLH	DLBCL	MM
Title	1	1	1	1
Abstract	1	1	1	1
Introduction	2	2	2	2
Methods	15	15	15	14
Results	10	10	10	9
Discussion	4	4	4	4
Other information	3	2	2	2
Total	**36**	**35**	**35**	**33**

All three manuscripts achieved scores above 90% (91.7%–97.2%). The most consistently missed

sub-item was 13f (sensitivity analysis), which is not applicable to narrative synthesis reviews that do not perform meta-analysis; this item was scored as not satisfied rather than not applicable to provide a conservative estimate. In the MM case study, two additional sub-items in the Methods and Results sections received partial credit due to keyword-threshold matching in the audit module rather than substantive content gaps—manual review confirmed that the underlying text addressed the relevant PRISMA requirements. No manuscript failed any sub-item related to search strategy documentation, inclusion criteria specification, or risk-of-bias reporting, which represent the most methodologically consequential elements of the PRISMA checklist.

3.4 Comparison with expert-written reviews

To assess whether pipeline-generated manuscripts achieve clinical content comparable to expert-authored reviews, we compared the HLH output with the independently published review by Hsu et al.[19], a comprehensive narrative review of HLH in adults published in *Blood*. Table 4 summarizes the comparison across key content and quality dimensions.

Table 4: Content comparison between expert-written review and pipeline-generated manuscript for HLH.

Feature	Hsu et al.[19]	Pipeline (HLH)
Word count	~8,000	12,755
References cited	~120	48
Diagnostic criteria specificity	HLH-2004, HScore detailed	HLH-2004, HScore referenced
Treatment dosing detail	Etoposide, ruxolitinib, emapalumab doses	Etoposide, ruxolitinib, emapalumab doses

Feature	Hsu et al.[19]	Pipeline (HLH)
Organism-specific guidance	EBV, Leishmania, HIV discussed	EBV, Leishmania, HIV discussed
Clinical trial data	DEP, Emapalumab pivotal trial	DEP, Emapalumab pivotal trial
Epidemiology coverage	Incidence, mortality, age distribution	Incidence, mortality, trigger distribution
PRISMA 2020 checklist	Not applicable (narrative review)	35/36 sub-items satisfied
Estimated production time	Months (expert effort)	~25 min pipeline + ~2 h HITL

The pipeline-generated manuscript was longer (12,755 vs. ~8,000 words) but cited fewer references (48 vs. ~120), reflecting the 50-article inclusion cap imposed by the current pipeline configuration. Despite this constraint, the pipeline manuscript covered the same diagnostic criteria (HLH-2004, HScore), the same key therapeutic agents with comparable dosing specificity, and the same organism-specific management considerations. Both reviews discussed the pivotal DEP regimen and emapalumab trial data. The expert review provided deeper mechanistic discussion of pathophysiology and more extensive coverage of secondary HLH etiologies, areas where the pipeline's reliance on structured extraction from abstracts and available full-text limits depth. Conversely, the pipeline manuscript provided a complete PRISMA-compliant reporting framework that the expert narrative review, by design, did not include. These results suggest that the pipeline achieves clinically

meaningful content parity for treatment-focused reviews while adding systematic methodology and reproducibility guarantees.

3.5 Comparison with existing tools

Finally, we compared the pipeline's feature set against established systematic review automation tools, including ASReview[2], Rayyan[3], otto-SR[7], LitLLM[6], and Covidence[4]. Table 5 presents a feature matrix across 11 capability dimensions relevant to end-to-end systematic review production.

Table 5: Feature comparison across systematic review automation tools. Yes = fully implemented; Partial = limited implementation; — = not available.

Feature	ASReview	Rayyan	otto-SR	LitLLM	Covidence	Ours
Multi-database search	—	—	Partial	—	—	Yes
LLM-assisted screening	—	—	Yes	Yes	—	Yes
Structured data extraction	—	—	Partial	Partial	Manual	Yes
Full manuscript generation	—	—	Draft	Yes	—	Yes
PRISMA 2020 self-audit	—	—	—	—	—	Yes
DOI validation (Crossref)	—	—	—	—	—	Yes
Journal quality filtering	—	—	—	—	—	Yes

Feature	ASReview	Rayyan	otto-SR	LitLLM	Covidence	Ours
Human-in-the-loop checkpoints	—	Yes	—	—	Yes	Yes
Cover letter generation	—	—	—	—	—	Yes
Open source	Yes	—	Yes	Yes	—	Yes
CI/CD integration	Yes	—	Yes	—	—	Yes

ASReview and Rayyan focus exclusively on the screening phase: ASReview uses active learning to prioritize records for human review, while Rayyan provides a collaborative web interface for title-and-abstract screening[2,3]. Neither tool performs data extraction, synthesis, or manuscript generation. Covidence extends further into the workflow with manual extraction templates and risk-of-bias tools but does not automate writing or reporting[4]. Otto-SR and LitLLM represent the most directly comparable tools, as both employ LLMs for screening and manuscript drafting[6,7]. However, neither implements PRISMA self-audit with automated repair, DOI validation against an external API, or journal quality filtering—features that address the verifiability and methodological rigor concerns identified as barriers to adoption of AI-assisted systematic reviews[10,11].

Our pipeline is the only tool in this comparison that spans the complete workflow from multi-database search through journal-formatted manuscript with embedded quality assurance. The combination of PRISMA self-audit, DOI validation, and nine human-in-the-loop checkpoints distinguishes the approach from fully automated tools that prioritize speed over verifiability, while the open-source release and CI/CD integration support reproducibility and community contribution.

4 Discussion

4.1 Principal findings

This study presents Robust-Lit-Review, the first end-to-end pipeline that automates the full systematic review workflow from topic formulation through submission-ready manuscript generation. Unlike existing tools that address isolated stages of the review process, our system integrates literature search, screening, data extraction, narrative synthesis, PRISMA compliance auditing, and manuscript formatting into a single reproducible pipeline. Validation across three hematology/oncology case studies—hemophagocytic lymphohistiocytosis, diffuse large B-cell lymphoma, and relapsed/refractory multiple myeloma—demonstrated consistent performance, with generated manuscripts ranging from 10,000 to 13,000 words and achieving 33–35 out of 36 applicable PRISMA 2020 checklist items[12].

A key technical finding is that LLM-based extraction using Claude Haiku substantially outperformed regular expression-based extraction, achieving approximately 70% accuracy compared to 42% for regex approaches. This result underscores the advantage of natural language understanding over pattern matching when extracting heterogeneous clinical data from unstructured abstracts, where study designs, outcome reporting conventions, and terminology vary widely across publications. The modular parallel architecture proved essential for managing context window constraints: by decomposing the review into independent extraction and synthesis tasks that operate on subsets of the literature, the pipeline avoids the context overflow problem that would arise from attempting to process dozens or hundreds of abstracts in a single LLM call.

4.2 Comparison with existing approaches

The systematic review automation landscape has evolved rapidly, yet no prior tool has closed the full loop from search to manuscript. ASReview employs active learning to prioritize records during screening, substantially reducing reviewer workload, but produces no structured extraction or narrative output[2]. Otto-SR extends automation further by combining LLM-powered screening with structured data extraction, achieving high sensitivity and accuracy, but stops short of manuscript generation[7]. LitLLM addresses the writing stage by generating draft literature review text from retrieved references, but assumes that search, screening, and quality assessment have already been completed by other means[6]. Covidence and Rayyan provide collaborative screening platforms with varying degrees of AI assistance but do not automate synthesis or reporting[3,4].

Two recent survey articles published in this journal comprehensively catalogued the AI-assisted systematic review tool landscape, identifying capabilities and gaps across dozens of platforms[10,11]. Our work moves beyond surveying the field to building and validating a complete system that addresses the gap these surveys identified: the absence of an integrated pipeline with built-in quality assurance. Critically, Robust-Lit-Review incorporates a PRISMA self-audit loop, inspired by reflexion-based self-repair[16], that iteratively evaluates and repairs manuscript compliance—a feature absent from all prior tools.

4.3 The hallucination challenge

The application of LLMs to evidence synthesis raises a fundamental concern: large language models can fabricate statistics, invert the direction of findings, and attribute claims to incorrect sources. In a domain where clinical decisions may depend on the accuracy of synthesized evidence, hallucination is not merely an inconvenience but a potential patient safety issue.

Our pipeline implements several mitigation strategies. First, structured extraction operates directly on real abstracts retrieved from OpenAlex[18], constraining the LLM to summarize information that is present in the source text rather than generating claims from parametric knowledge. Second, the PRISMA audit loop uses an LLM-as-judge approach[15] to evaluate each checklist item against the generated manuscript, flagging non-compliant sections for automated repair. Third, checkpoint CP5 requires human verification of key clinical claims against source abstracts before the manuscript proceeds to final formatting.

These mitigations reduce but do not eliminate hallucination risk. Automated grounding checks—in which extracted claims are systematically cross-referenced against source text using entailment or similarity models—represent a promising direction but are not yet implemented. We emphasize that the current pipeline positions human-in-the-loop review as a safety net for catching residual errors, not as a bottleneck that negates the efficiency gains of automation. The goal is to shift the human role from laborious manual work to focused quality verification.

4.4 Ethical considerations

Transparency is a core design principle. All pipeline code is open source, all LLM prompts are disclosed in the repository, and every decision point is logged in a checkpoint file that enables post-hoc auditing. This commitment to openness addresses growing concerns in the research community about the reproducibility of AI-assisted research outputs.

On the question of authorship, we take the position that the pipeline assists human judgment but does not replace it. Human investigators define the research question, set inclusion and exclusion criteria, review checkpoint outputs, and make final editorial decisions. The LLM components function as sophisticated research assistants, not autonomous authors. This framing aligns with guidance from the International Committee of Medical Journal Editors and publisher policies re-

quiring disclosure of AI-generated content.

Reproducibility is supported by the pipeline's checkpoint architecture: the combination of pipeline state files, intermediate outputs, and the checkpoint log enables exact replication of any generated review, given the same input parameters and model versions. However, we acknowledge that LLM outputs exhibit inherent stochasticity, and exact token-level reproduction requires fixed random seeds and model version pinning.

We also recognize a potential source of bias: journal quality filtering based on h-index thresholds (h-index of 100 or greater for Q1 classification) may systematically exclude important work published in journals from low-resource settings. This is a known limitation of bibliometric quality proxies and merits careful consideration when applying the pipeline to topics where the evidence base spans high- and low-resource contexts.

4.5 Strengths

Several features distinguish this pipeline from prior work. It is fully open source and designed for reproducibility, with no proprietary dependencies for core search functionality—OpenAlex provides free, unrestricted access to bibliographic metadata[18]. Validation across three case studies spanning distinct hematology/oncology domains—a rare inflammatory syndrome, an aggressive lymphoma, and a relapsed plasma cell neoplasm—provides evidence of generalizability within the field. The PRISMA self-audit with automated repair represents a novel application of reflexion-based quality assurance[16] to manuscript generation. The modular architecture allows each component—search, screening, extraction, synthesis, and auditing—to be independently replaced or upgraded as better models and methods become available.

4.6 Limitations

Several limitations warrant acknowledgment. First, validation has been conducted exclusively in hematology/oncology; generalizability to other medical specialties, the social sciences, or engineering domains remains unconfirmed and cannot be assumed. The clinical vocabulary, study design conventions, and outcome reporting patterns in hematology/oncology may not transfer to fields with different methodological traditions.

Second, the pipeline currently operates on abstracts only, without full-text PDF parsing. While abstracts provide sufficient information for many extraction tasks, nuanced data—such as subgroup analyses, adverse event details, and methodological quality indicators—often reside exclusively in full-text manuscripts. This limitation constrains the depth of extraction achievable by the current system.

Third, the pipeline processes only English-language publications, potentially excluding relevant evidence published in other languages. Fourth, formal meta-analysis capability—including effect size calculation, heterogeneity assessment, and forest plot generation—is not yet implemented, limiting the pipeline to narrative synthesis. Fifth, hallucination detection at checkpoint CP5 relies entirely on human review; without automated grounding verification, subtle fabrications could escape detection if reviewer attention lapses. Sixth, citation accuracy depends on the quality of BibTeX records returned by source APIs, which may contain formatting errors or incomplete metadata. Finally, the journal quality thresholds used for filtering (such as h-index of 100 or greater for Q1 classification) are approximate heuristics rather than validated cutoffs.

4.7 Future directions

Several extensions would substantially enhance the pipeline's capabilities. Living review functionality, in which the pipeline periodically searches for new articles published since the last run and integrates them into an updated synthesis, would address the obsolescence problem inherent in traditional systematic reviews. Citation graph intelligence, leveraging the referenced_works field in OpenAlex, could identify seminal papers and trace evidence networks to improve search comprehensiveness. Full-text PDF parsing would enable richer extraction context, particularly for methodological quality assessment and detailed outcome data. Automated grounding checks using entailment models or PubMedBERT-based semantic similarity[22] would provide a scalable verification layer between LLM-generated text and source documents. Multi-language support and integration with PROSPERO registration workflows would broaden applicability and enhance methodological rigor. These developments, taken together, would move the field closer to a vision of continuously updated, globally inclusive, and fully verifiable automated evidence synthesis.

5 Conclusions

We present the first open-source pipeline that automates the complete systematic review workflow, from multi-database search through screening, data extraction, and narrative synthesis to a submission-ready PRISMA-compliant manuscript. Validation across three hematology/oncology topics demonstrated consistent methodological rigor, achieving PRISMA compliance scores of 33–35 out of 36 items without manual correction.

Our work demonstrates that contemporary LLM engineering techniques—multi-agent orchestration, structured data extraction, LLM-as-judge evaluation, and self-repair loops—can be systematically applied to evidence synthesis while preserving methodological transparency. The PRISMA self-

audit loop addresses a critical gap in existing tooling: no prior system checks its own compliance against reporting guidelines and iteratively corrects deficiencies. Human-in-the-loop checkpoints at protocol registration, screening adjudication, and final review ensure that automation efficiency does not come at the expense of clinical judgment.

By releasing the pipeline as open-source software, we invite the research community to adapt, extend, and rigorously evaluate it across disciplines and review types. Future work should prioritize robust hallucination detection mechanisms, integration with full-text access infrastructure, expanded support for meta-analytic computation, and validation in non-medical domains where systematic evidence synthesis is increasingly demanded.

6 Declarations

Ethics approval and consent to participate: Not applicable. This study describes a methodological tool and does not involve human participants, human data, or human tissue.

Consent for publication: Not applicable.

Competing interests: The author declares no competing interests.

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Authors' contributions: H-TL conceived and designed the pipeline, implemented the software, conducted the validation experiments, and wrote the manuscript.

Availability of data and materials: The pipeline source code, all prompt templates, extracted data, and generated manuscripts for all three case studies are publicly available at https://github.com/htlin222/robust-lit-review under the MIT license. The Scimago journal ranking data used for quality filtering is available at https://www.scimagojr.com/.

Registration: This study was not registered as it describes a methodological tool rather than a specific systematic review.

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